release of ADH from the posterior pituitary. The resulting increase in water reabsorption in the collecting duct concentrates urine, reduces urine volume, and lowers blood osmolarity back toward the normal range. As the osmolarity of the blood falls, a negative-feedback mechanism reduces the osmoreceptor cell activity in the hypothalamus, and ADH secretion is reduced.

What happens if, instead of ingesting salt or sweating profusely, you drink a large amount of water? Blood osmolarity falls below the normal range, causing a drop in ADH secretion to a very low level. The resulting decrease in permeability of the collecting ducts reduces water reabsorption, resulting in discharge of large volumes of dilute urine.

Contrary to common belief, caffeinated drinks increase urine production to no greater degree than water of comparable volume: Numerous studies of coffee and tea drinkers have found no diuretic effect for caffeine.

Blood osmolarity, ADH release, and water reabsorption in the kidney are normally linked in a feedback circuit that contributes to homeostasis. Anything that disrupts this circuit can interfere with water balance. For example, alcohol inhibits ADH release, leading to excessive urinary water loss and dehydration (which may cause some of the symptoms of a hangover).

Mutations that prevent ADH production or that inactivate the ADH receptor gene disrupt homeostasis by blocking the insertion of additional aquaporin channels in the collecting duct membrane. The resulting disorder can cause severe dehydration and solute imbalance due to production of copious dilute urine. These symptoms give the disorder its name: *diabetes insipidus* (from the Greek for "to pass through" and "having no flavor"). Could mutations in an aquaporin gene have a similar effect? **Figure 44.20** describes an experimental approach that addressed this question.

The Renin-Angiotensin-Aldosterone System

The release of ADH is a response to an increase in blood osmolarity, as when the body is dehydrated from excessive water loss or inadequate water intake. However, an excessive loss of both salt and body fluids—caused, for example, by a major wound or severe diarrhea—will reduce blood volume *without* increasing osmolarity. Given that this will not affect ADH release, how does the body respond? It turns out that an endocrine circuit called the **renin-angiotensin-aldosterone system (RAAS)** also regulates kidney function. The RAAS responds to the drop in blood volume and pressure by increasing water and Na^+ reabsorption.

The RAAS involves the **juxtaglomerular apparatus (JGA)**, a specialized tissue consisting of cells of and around the afferent arteriole, which supplies blood to the glomerulus. When blood pressure or volume drops in the afferent

▼ Figure 44.20 Inquiry

Can aquaporin mutations cause diabetes?

Experiment Researchers studied a patient with diabetes insipidus who had a normal ADH receptor gene but two mutant alleles (A and B) of the aquaporin-2 gene. The resulting changes are shown below in an alignment of protein sequences that includes other species.

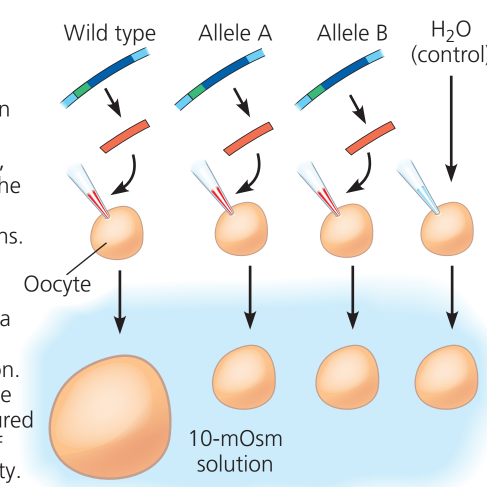
Source of Aquaporin-2 Gene Sequence	Amino Acids 183–191* in Encoded Protein	Amino Acids 212–220* in Encoded Protein
Frog (*Xenopus laevis*)	MNPARSFAP	GIFASLIYN
Lizard (*Anolis carolinensis*)	MNPARSFGP	AVVASLLYN
Chicken (*Gallus gallus*)	MNPARSFAP	AAAASIIYN
Human (*Homo sapiens*)	MNPARSLAP	AILGSLLYN
Conserved residues	MNPARS-P	-S-YN
Patient's gene: allele A	MNPACSLAP	AILGSLLYN
Patient's gene: allele B	MNPARSLAP	AILGPLLYN

*The numbering is based on the human aquaporin-2 protein sequence.

Each mutation changed the protein sequence at a highly conserved position. To test the hypothesis that the changes affect function, researchers used frog oocytes, cells that will express foreign messenger RNA and can be readily collected from adult female frogs.

1 Messenger RNA transcribed from wild-type and mutant aquaporin genes is injected into frog oocytes, where it directs the synthesis of aquaporin proteins.

2 The oocytes are transferred from a 200-mOsm to a 10-mOsm solution. The rate of oocyte swelling is measured as an indicator of water permeability.



Results

Source of Injected mRNA	Rate of Swelling (µm/sec)
Human wild type	196
Patient's allele A	17
Patient's allele B	18
None (H_2O control)	20

Conclusion Because each mutation renders aquaporin inactive as a water channel, researchers concluded that these mutations cause the disorder common to the patients.

Data from P. M. Deen et al., Requirement of human renal water channel aquaporin-2 for vasopressin-dependent concentration of urine, *Science* 264:92–95 (1994).

WHAT IF? *If you measured ADH levels in patients with ADH receptor mutations and in patients with aquaporin mutations, what would you expect to find, compared with wild-type subjects?*